Example 9.21

Calculate $\lim_{x\to 0} \frac{1}{(x^2+x^3)}$.

Solution

One can tabulate values of x near 0 (from either side) and conclude $f(x)=\frac{1}{x^2+x^3}$ grows without bound and hence $f(x)\to\infty$ as $x\to 0$.

To calculate this limit without making a table, we first divide the numerator and denominator by x^2. This division can be done, since in the calculation of the limit $x\neq 0$ and hence $x^2\neq 0$. We can have

$$\lim_{x\to 0}\frac{1}{x^2+x^3} = \lim_{x\to 0}\frac{\frac{1}{x^2}}{\frac{x^2+x^3}{x^2}} = \frac{\lim_{x\to 0}\left(\frac{1}{x^2}\right)}{\lim_{x\to 0}(1+x)}.$$

Now $\frac{1}{x^2}\to\infty$ as $x\to 0$ and $\lim_{x\to 0}(1+x)=1$.

Thus the numerator grows without bound while the denominator approaches 1, implying that $\frac{1}{(x^2+x^3)}$ does tend to infinity.

This example illustrates how a difficult calculation can be greatly simplified by a few algebraic manipulations.

Example 9.22

Evaluate $\lim_{x\to 2}\frac{1}{(x-2)^3}$.

Solution

From the graph of $f(x)=\frac{1}{(x-2)^3}$,

clearly, $\frac{1}{(x-2)^3}\to-\infty$ as $x\to 2^-$ and

$\frac{1}{(x-2)^3}\to\infty$ as $x\to 2^+$.

Hence the limit does not exist.

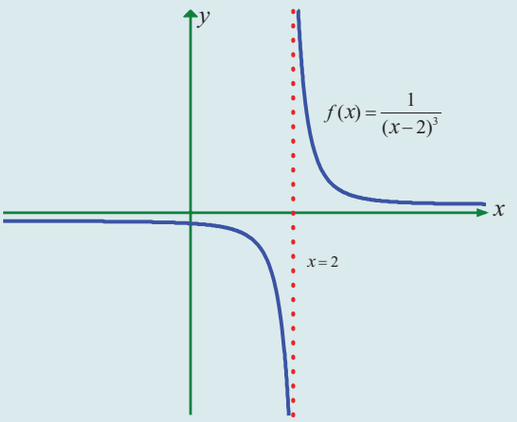


Fig. 9.24

In general

(i) If n is an even positive integer then

$\frac{1}{(x-a)^n}\to\infty$ as $x\to a$

$\frac{1}{(x-a)^n}\to\infty$ as $x\to a^-$

$\frac{1}{(x-a)^n}\to\infty$ as $x\to a^+$

(ii) If n is an odd positive integer, then

$\frac{1}{(x-a)^n}\to-\infty$ as $x\to a^-$

$\frac{1}{(x-a)^n}\to+\infty$ as $x\to a^+$.

The line $x=a$ becomes a vertical asymptote.

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